

SHOAIB SULTAN

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[LinkedIn](#) | [Live Portfolio](#)

EDUCATION

Lahore Garrison University, BS Computer Science

Nov 2022 - April 2026

- Cumulative GPA: 3.55/4.0

- Relevant Coursework: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Database Management Systems (DBMS), Operating Systems (OS), Software Engineering, Computer Networks, and Problem Solving.

PROJECTS

RAGSphere – Enterprise AI Document Intelligence Platform | FastAPI / React 19 / FAISS / BM25 / Gemini / Docker / Tailwind CSS |

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- **[Enterprise RAG System]** Developed a production-grade Retrieval-Augmented Generation (RAG) platform featuring Hybrid Dense-Sparse Retrieval, FAISS Vector Search, BM25 Lexical Matching, and intelligent document-grounded response generation.
- **[Advanced Document Processing Pipeline]** Built an end-to-end PDF ingestion workflow including text extraction, chunking, metadata generation, vector embedding creation, document indexing, and semantic retrieval for enterprise knowledge management.
- **[Context-Aware Conversational AI]** Implemented conversation-history-aware query rewriting, semantic expansion, and retrieval re-ranking to improve answer relevance, contextual understanding, and multi-turn chat accuracy.
- **[Full-Stack AI SaaS Deployment]** Engineered a scalable React + FastAPI architecture with Docker-based deployment, real-time retrieval diagnostics, cloud hosting support (Railway/Vercel), and production-ready API integrations.

Person and Hivis Detection System | FastAPI / React / YOLOv8 / ONNX / OpenCV / WebSockets

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- **[Real-Time AI Detection System]** Built a production-level vision system using YOLOv8 ONNX Runtime to detect people and hi-vis safety jackets with high accuracy and low latency.
- **[Optimized Inference Pipeline]** Designed a high-performance OpenCV + ONNX processing pipeline with preprocessing, NMS, and real-time frame analysis for smooth live detection.
- **[Full-Stack Real-Time Dashboard]** Developed a modern React + Vite frontend with glassmorphism UI, Tailwind CSS, and WebSocket-based live visualization for real-time AI results.

Histopathology Cancer Classification System | ResNet-18, Vision Transformer (ViT), ONNX, OpenCV

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- **[Multi-Class Cancer Classification]** Developed a deep learning system to classify histopathology images into 5 categories: Colon Adenocarcinoma, Benign Colon Tissue, Lung Adenocarcinoma, Benign Lung Tissue, and Lung Squamous Cell Carcinoma using ResNet-18 and Vision Transformer (ViT).
- **[Deep Learning Architecture – ResNet + ViT]** Applied transfer learning with ResNet-18 for strong baseline performance and Vision Transformer (ViT) for enhanced global feature extraction in complex histopathology patterns.
- **[Medical Image Pipeline & Augmentation]** Built a robust preprocessing pipeline using OpenCV including resizing, normalization, and data augmentation to improve model generalization across varied medical image samples.
- **[ONNX Deployment & Performance]** Exported model to ONNX for fast inference and production readiness, achieving ~92–96% accuracy, demonstrating strong potential for AI-assisted pathology workflows.

Associate AI/ML Engineer

July 2026– Present

@ AB Ark Solution

- Architected and optimized end-to-end AI/ML solutions by leveraging advanced Machine Learning, Deep Learning, Computer Vision, and NLP techniques to solve complex business problems.
- Engineered scalable data pipelines, performed feature engineering, model optimization, hyperparameter tuning, and rigorous performance evaluation to maximize model accuracy and reliability.
- Collaborated with cross-functional engineering teams to deploy production-ready AI models, integrate REST APIs, and automate inference workflows while ensuring scalability, efficiency, and maintainability.

Intern AI/ML

Nov 2025 – Feb 2026

@ Devsinc

- Built and experimented with machine learning and deep learning models for real-world problem solving.
- Worked on data preprocessing, feature engineering, and model training using Python and ML libraries.
- Assisted in developing AI/ML pipelines and improving model performance through evaluation and tuning.
- Collaborated with the team to understand production-level AI workflows and deployment concepts.

Intern AI/ML

July 2025 – Sept 2025

@ Digital Portal Official.

- Developed and deployed deep learning models for computer vision using PyTorch and TensorFlow.
- Designed efficient data pipelines for large-scale training and model optimization.
- Delivered end-to-end AI solutions by integrating models into production environments.

LEADERSHIP ACTIVITIES

Campus Director at Lahore Garrison University

Dec 2023 – Feb 2024

@ Hult-Prize

- Led outreach and advertising efforts, successfully attracting innovative project teams.
- Managed and hosted a major event, concluding with awarding prizes and promotion of winners.

AWARDS AND ACHIEVEMENTS

- Recognized on the VC List for achieving a 3.95 CGPA.

- Secured 2nd position in CP Category and received cash prize & certificate by Tech Gala Hackathon.

SKILLS / TECHNOLOGIES

Languages: Python for Automation & Scripting, PyTorch & TensorFlow for Deep Learning, OpenCV for Computer Vision, Generative AI (LLMs & Diffusion Models), Scikit-learn for Machine Learning, Pandas & NumPy for Data Processing

Tools: Tools: Git, GitHub, Docker, Visual Studio Code, PyCharm, Jupyter Notebook, Anaconda, Postman, Linux (Ubuntu), GitHub Actions, CI/CD